

Software Design Document

Bone fracture detection using AI



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Summary

Bone Fracture Detection Using AI is an intelligent healthcare system that uses Artificial Intelligence and deep learning techniques to analyze X-ray images and detect bone fractures automatically. The system helps doctors and medical staff identify fractures quickly and accurately, reducing manual effort and diagnosis time.

It uses image processing algorithms such as Convolutional Neural Networks (CNN) to examine medical images and generate fracture detection results. The application also provides features such as X-ray image upload, real-time detection, fracture report generation, notifications, and secure medical data handling. This AI-based solution improves healthcare efficiency, enhances diagnostic accuracy, and supports faster medical decision-making.

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1. Application Architecture

The Bone Fracture Detection Using AI system follows the MVVM (Model-View-ViewModel) architecture to ensure scalability, maintainability, and efficient performance. The architecture separates the user interface, business logic, and data handling into different layers. This structure helps the application perform AI-based fracture detection accurately and efficiently.

1.1 Architecture Components

The application architecture consists of Models, Views, and View Models. These components work together to process medical images, display fracture detection results, and manage communication between the user interface and AI system.

1.1.1 Models

Models manage the application data and business logic. They store patient details, uploaded X-ray images, AI detection results, reports, and notifications. Models also communicate with databases and AI algorithms for image processing and result generation.

1.1.2 Views

Views represent the graphical user interface of the application. They display screens such as login, image upload, fracture results, notifications, and medical reports. Views are responsible for presenting information and collecting user input.

1.1.3 View Models

View Models act as a bridge between Models and Views. They process user actions, manage data flow, and update the interface whenever information changes. View Models also handle communication between the AI detection model and the user interface.

1.2 Application Architecture Diagram

The Application Architecture Diagram visually represents the overall structure of the Bone Fracture Detection system. It illustrates the interaction between the mobile application, AI processing model, cloud database, APIs, and user interface components.

1.3 APIs and Libraries

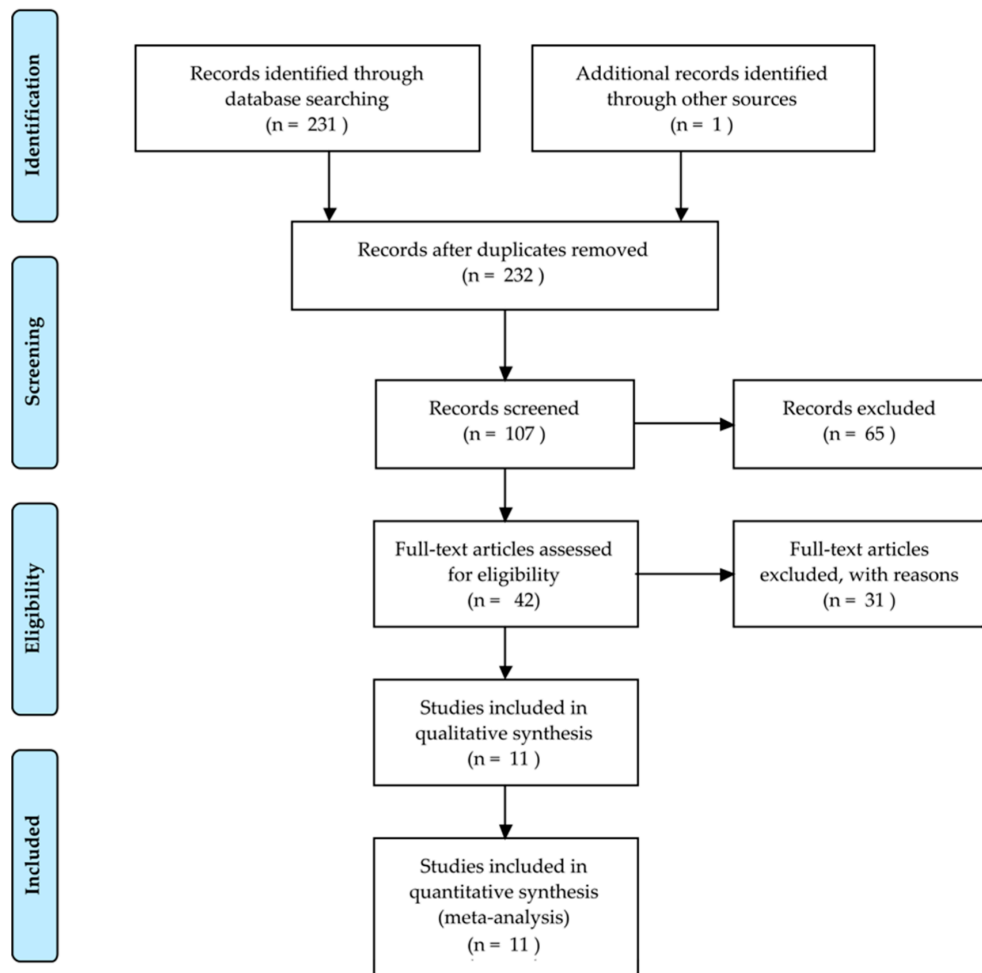
The application uses different APIs and libraries to support AI processing, image handling, authentication, and database management.

Technologies and libraries include:

- TensorFlow / TensorFlow Lite
- OpenCV
- Firebase Authentication
- Firebase Firestore
- Retrofit API
- Glide/Picasso
- Python Flask API

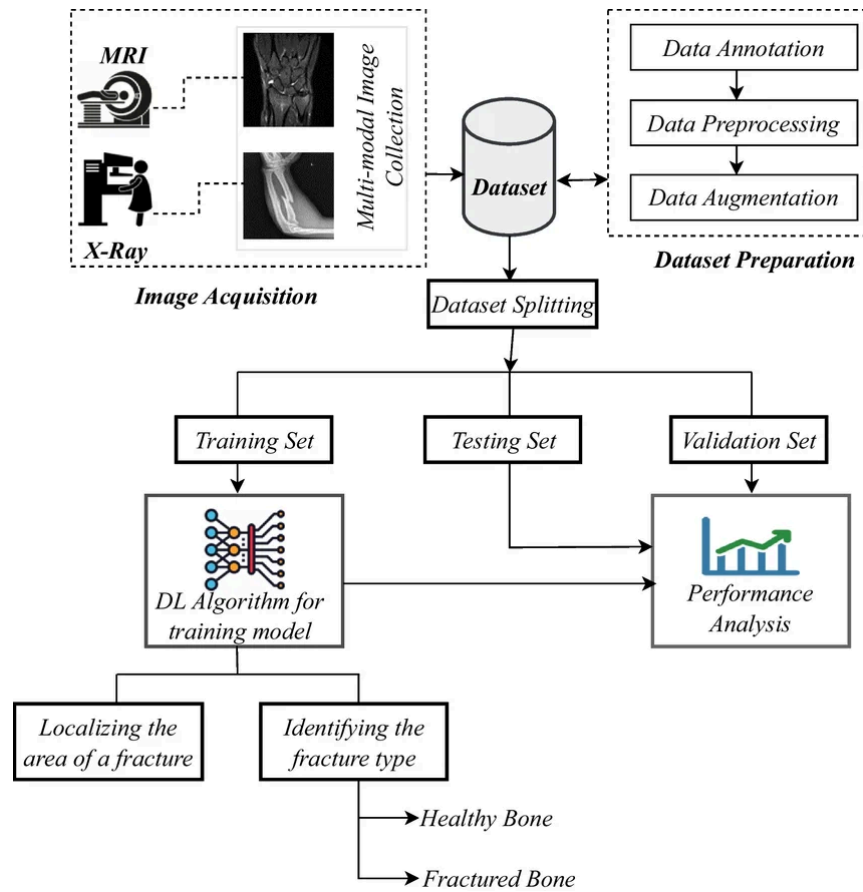
1.4 UML Diagrams

UML diagrams are used to visually explain the workflows, structures, and interactions within the Bone Fracture Detection system. These diagrams help developers understand system functionality and data flow.



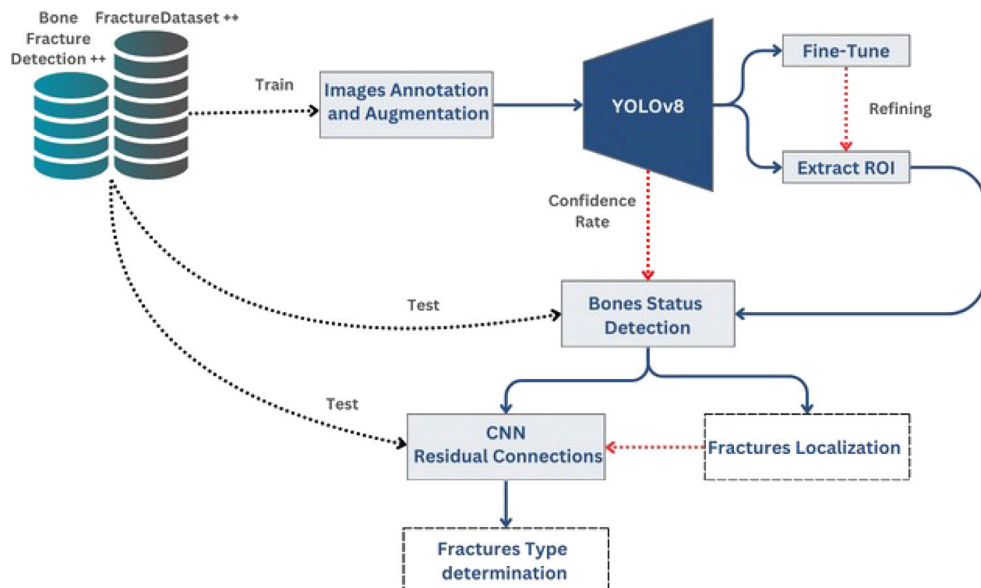
1.4.1 Use Case Diagram

The Use Case Diagram shows the interaction between users, doctors, and the AI system. It includes activities such as login, X-ray upload, fracture detection, report generation, and result viewing.



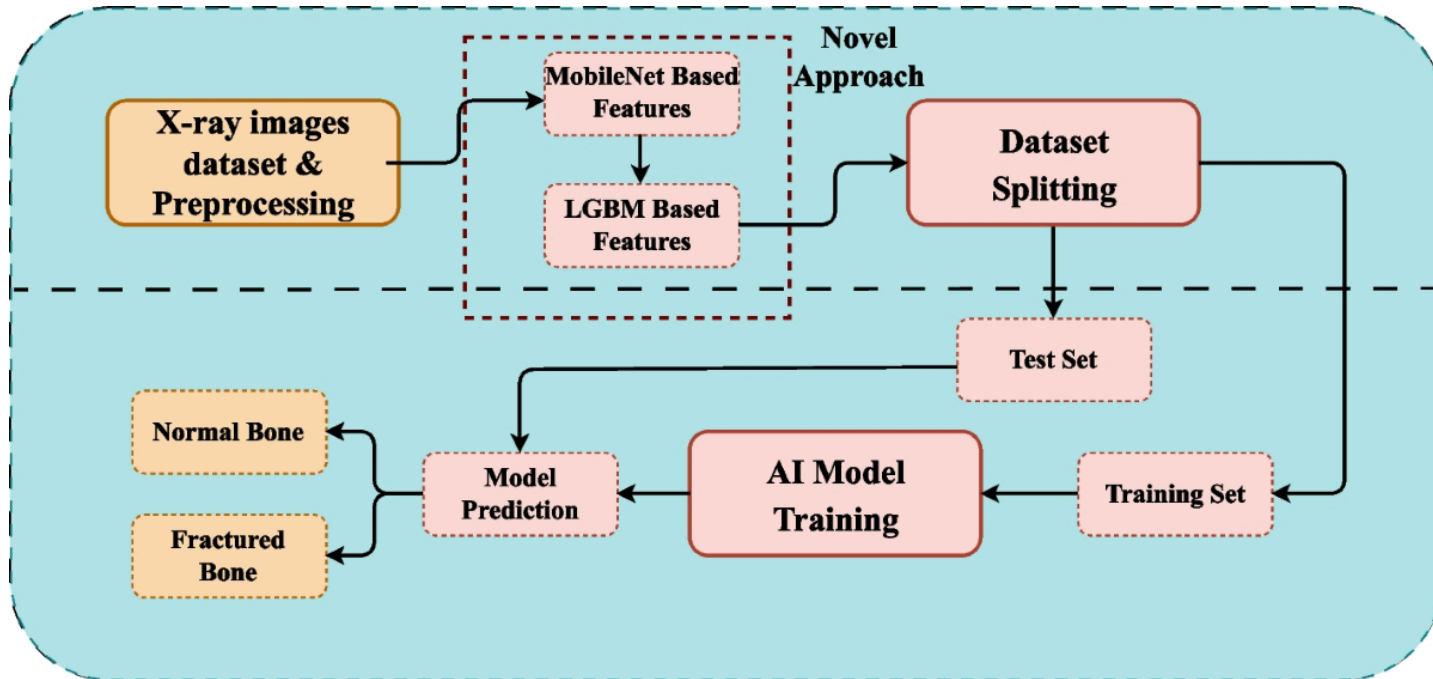
1.4.2 Class Diagram

The Class Diagram represents the classes used in the system such as User, Patient, X-ray Image, Detection Result, Report, and Notification. It also shows the relationships between these classes.



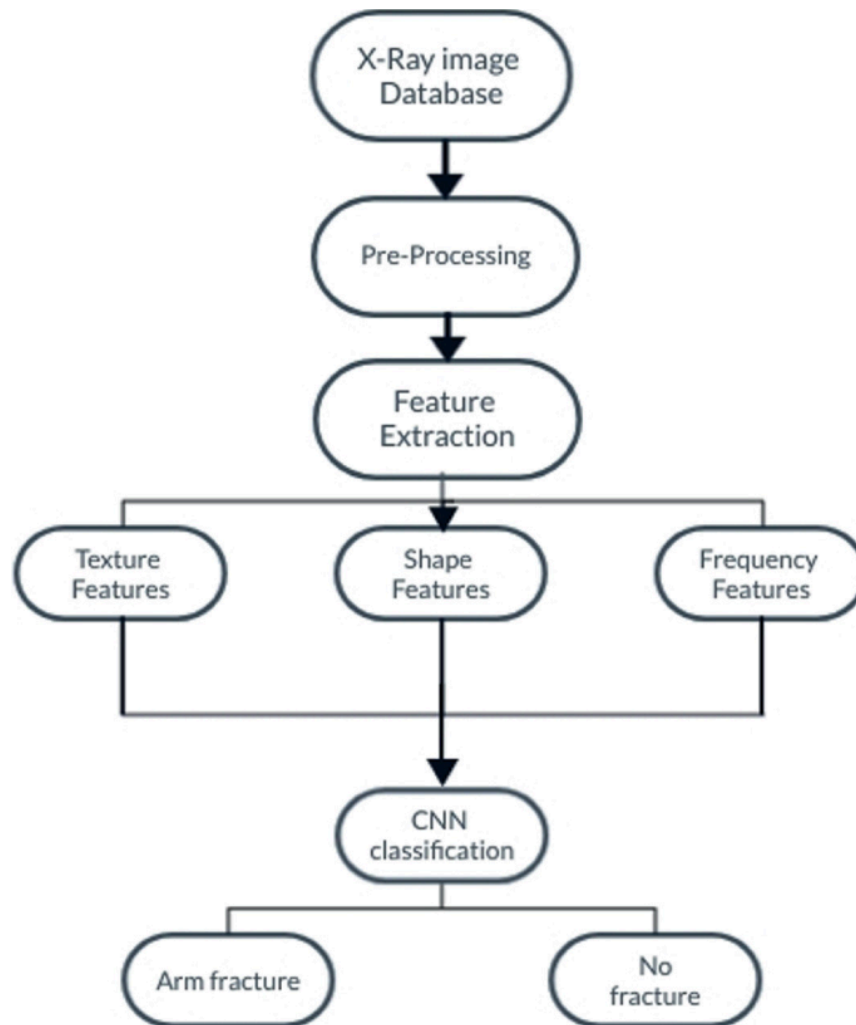
1.4.3 Sequence Diagram

The Sequence Diagram explains the step-by-step communication between the user, application interface, AI model, and database during the fracture detection process.



1.4.4 Activity Diagram

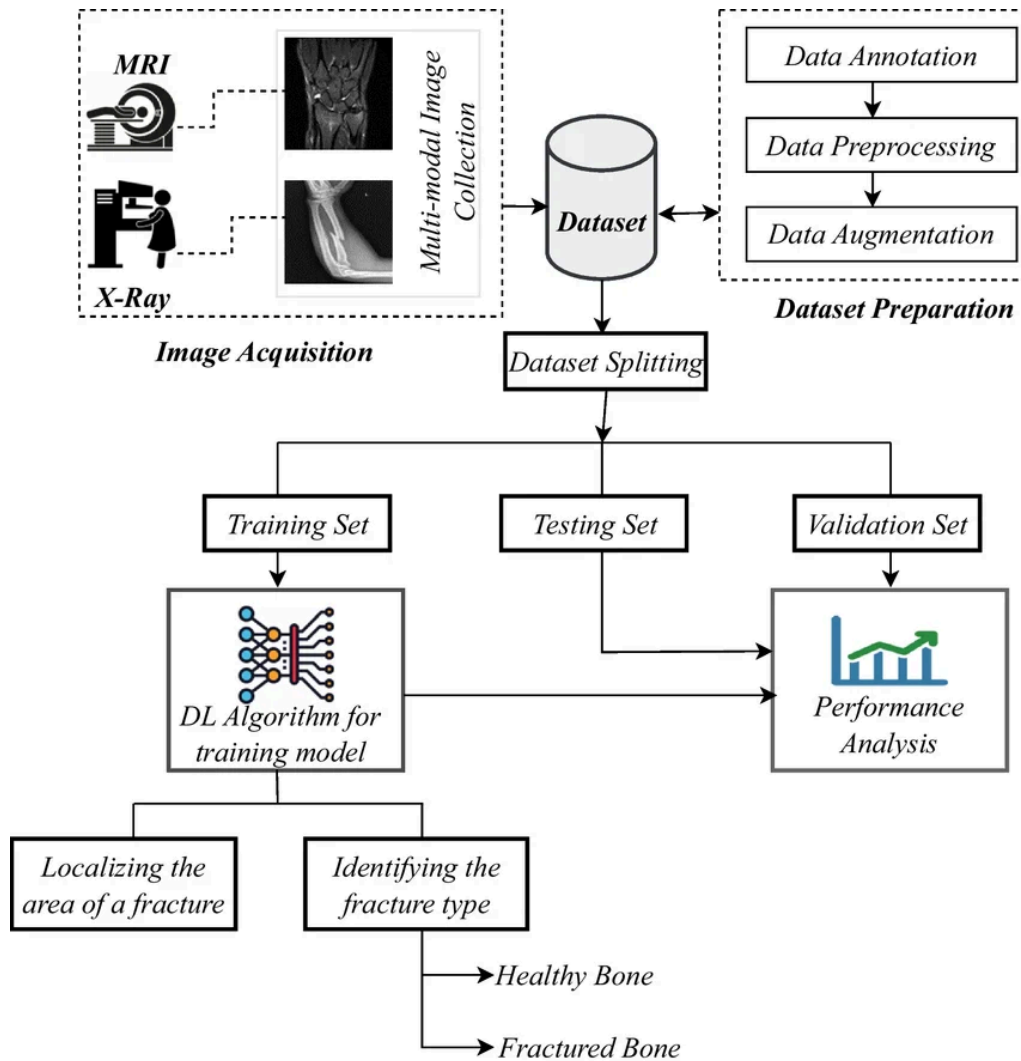
The Activity Diagram illustrates the workflow of the application including login, image upload, AI analysis, result generation, and report viewing processes.



1.4.5 Deployment Diagram

The Deployment Diagram represents the physical deployment of the application including mobile devices, cloud servers, AI processing systems, APIs, and databases connected through the internet.

1.5 Coding Conventions



Coding conventions are followed to maintain clean, readable, and maintainable code. Proper indentation, meaningful variable names, comments, modular programming, and reusable methods are used throughout the application.

1.6 Algorithms and Technologies

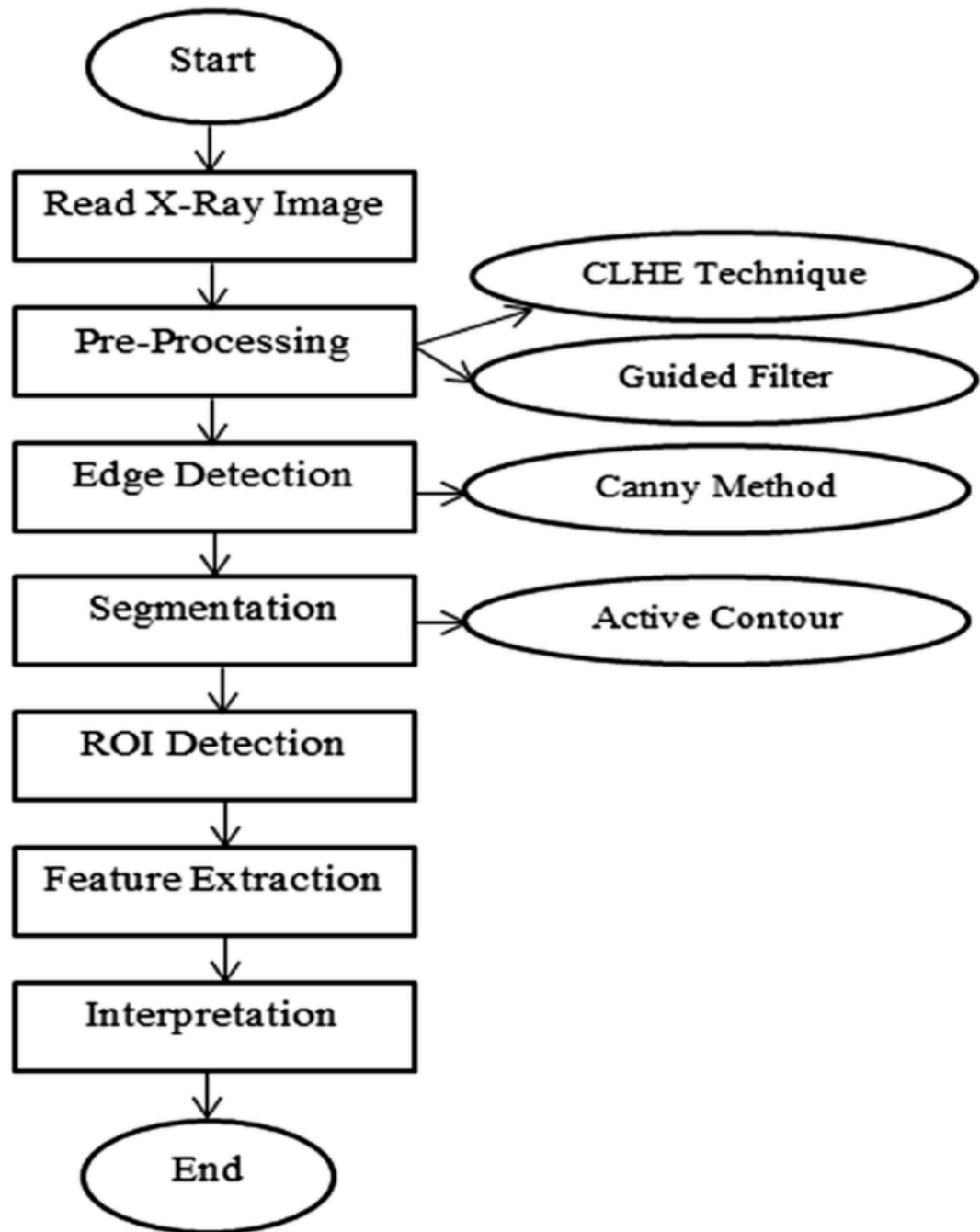
The Bone Fracture Detection system uses AI and deep learning algorithms to analyze X-ray images and identify fractures accurately.

Technologies and algorithms include:

- Convolutional Neural Network (CNN)
- Image Classification Algorithms
- TensorFlow AI Model
- OpenCV Image Processing
- Firebase Cloud Services

1.7 Bone Fracture Detection Features

The application provides advanced AI-based features that help users and medical professionals detect fractures quickly and efficiently.



1.7.1 X-ray Image Upload

Users can upload X-ray images from their gallery or camera for fracture analysis. The system supports fast and secure image uploading.

- Allows users to upload X-ray images from the gallery or camera.
- Supports quick and secure image uploading.
- Ensures images are safely stored for analysis.
- Provides the first step for accurate fracture detection.

1.7.2 AI-Based Fracture Detection

The AI model processes uploaded X-ray images and detects fractures automatically using deep learning techniques.

1.7.3 Fracture Report Generation

After image analysis, the system generates a detailed fracture report containing fracture probability and detection information.

- Generates a detailed fracture report after image analysis.
- Includes fracture probability and detection results.
- Provides clear information about the identified fracture.
- Helps doctors and users review the diagnosis efficiently.

1.7.4 Real-Time Detection Results

Users can view fracture detection results instantly after image processing. The system highlights possible fracture areas for better understanding.

- Displays fracture detection results immediately after image processing.
- Highlights suspected fracture areas on the uploaded image.
- Helps users quickly understand the diagnosis.
- Improves accuracy and speeds up medical decision-making.

1.7.5 Doctor Recommendation System

The application may recommend medical specialists or doctors based on fracture severity and AI detection results.

1.7.6 Notifications and Alerts

The system sends notifications related to image processing, fracture results, and report generation updates.

- Sends updates about image processing status.
- Notifies users when fracture detection results are available.
- Alerts users when the medical report is generated.

2. Data Model Schema

The Data Model Schema defines how application data is organized and stored. It manages patient information, X-ray images, AI detection results, reports, and notifications.

2.1 Main Entities

Main entities represent the core components of the database used in the Bone Fracture Detection system.

2.1.1 User

The User entity stores login credentials and profile information such as username, email, password, and user role.

2.1.2 Patient

The Patient entity stores patient details including patient ID, name, age, gender, and medical history.

2.1.3 X-ray Image

The X-ray Image entity stores uploaded medical images and image-related information such as upload date and image path.

2.1.4 AI Detection Result

This entity stores fracture detection results generated by the AI model including prediction accuracy and identified fracture regions.

2.1.5 Fracture Report

The Fracture Report entity stores generated reports and medical analysis summaries related to fracture detection.

2.1.6 Notification

The Notification entity manages alerts and updates related to image processing, fracture reports, and medical recommendations.

2.2 Relationship Overview

The Relationship Overview explains how entities are connected within the database. A patient can upload multiple X-ray images, each image generates detection results, and reports are linked to patients and AI analysis outcomes.

- Shows how all database entities are connected in the system.
- A patient can upload multiple X-ray images for analysis.
- Each X-ray image produces corresponding detection results.
- Reports are generated based on AI analysis and linked back to the patient.

2.3 Design Consideration

Design considerations focus on creating a secure, scalable, and efficient AI-based medical system. Proper system planning ensures reliable performance and accurate fracture detection.

- Focuses on building a secure, scalable, and efficient AI-based medical system.
- Ensures reliable performance during image processing and fracture detection.
- Improves system accuracy through proper architecture and planning.
- Supports smooth and fast user experience for medical analysis.

2.3.1 ER Diagram

The Entity Relationship Diagram visually represents relationships between entities such as User, Patient, X-ray Image, Detection Result, Report, and Notification.

- The Entity Relationship Diagram shows the structure of the system.
- It represents relationships between entities like User, Patient, X-ray Image, Detection Result, Report, and Notification.
- Helps in understanding how data is connected and flows in the system.
- Provides a clear database design for implementation and development.

3. User Interface

The User Interface section describes the design and functionality of the application screens. The interface is designed to be simple, user-friendly, and responsive.

3.1 List of Screens

The application contains multiple screens for user interaction and fracture detection processes.

Screens include:

- Splash Screen
- Login Screen
- Registration Screen
- Home Screen
- X-ray Upload Screen
- Detection Result Screen
- Fracture Report Screen
- Notification Screen
- Profile Screen
- Admin Dashboard

3.2 User Interface Workflow

The User Interface Workflow explains how users interact with the system from login to fracture report generation.

3.2.1 User Login Workflow

This workflow explains the process of user authentication from entering login credentials to successful access to the application.

3.2.2 X-ray Upload Workflow

This workflow describes how users upload X-ray images for AI-based fracture analysis.

3.2.3 AI Detection Workflow

The AI Detection Workflow explains how the AI model processes X-ray images and generates fracture detection results.

3.2.4 Report Generation Workflow

This workflow illustrates the process of generating fracture reports after successful image analysis.

3.2.5 Notification Workflow

The Notification Workflow explains how users receive alerts and updates related to image processing and fracture reports.

4. Security Features

The Security Features section explains the methods used to protect user information, medical records, and application data.

4.1 User Authentication Security

The application uses secure login authentication and encrypted credentials to prevent unauthorized access.

4.2 Secure Medical Data Handling

Medical data such as X-ray images and fracture reports are securely stored and managed using protected cloud services.

4.3 Data Encryption

Data encryption techniques are used to secure sensitive information during storage and transmission.

5. Conclusion

Bone Fracture Detection Using AI is an intelligent healthcare solution that uses artificial intelligence and deep learning techniques to analyze X-ray images and detect fractures accurately. The system improves medical diagnosis efficiency, reduces manual workload, and provides faster fracture analysis and report generation.